Jonathan M. Fisher

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Objective

Mechanical Engineering major with hands-on experience in robotics and automation systems. Strong background in mechanical design, prototyping, and robotics, with practical skills in machining, pneumatics, PCB design, and embedded programming. Proven leadership and technical instruction experience through the Invention Studio and engineering internships. Seeking a Summer 2027 internship or co-op in robotics engineering, automation, manufacturing engineering, or mechanical systems design.

Education

Georgia Institute of Technology| Atlanta, GA — *August 2023 – Present*

Pursuing BS in Mechanical Engineering, GPA 3.77 — Expected Graduation, December 2026

Pursuing MS in Mechanical Engineering — Expected Graduation, December 2027

Skills

Mechanical & Prototyping: Machining (lathe, manual mill, band saw), problem solving, pneumatics, plumbing, soldering, wiring, PCB design, 3D printing (FDM & SLA), waterjet, laser cutting, woodworking, hand/power tools

Software & CAD: SolidWorks, Autodesk Inventor, KiCAD, Arduino IDE, Automation Direct Ladder Logic, Jands Vista, ProPresenter

Programming: Python, C++, C#, Java, JavaScript, SQL, HTML, TI Basic, ladder logic, analytical thinker

Hardware & Electronics: Arduino microcontrollers, electrical wiring, custom PCB fabrication, circuit design

Communication & Leadership: Technical instruction, team leadership/ management , public speaking, customer service

Languages: English (fluent), German (conversational)

Experience

Thermopore Material Company | Newnan, Ga — *May 2024 – August 2024, May 2025 – August 2025*

Engineering Co-op

- Machined parts and assembled machines from design drawings, incorporating manufacturing and design considerations.
- Gained hands-on experience with pneumatics, plumbing, field wiring, and PLC programming.
- Designed in SolidWorks and fabricated systems for a vibratory part escapement using 3D printing (FDM & SLA).
- Created a custom program to process McMaster-Carr CAD bolt files, reducing design engineer setup time by ~2 minutes per fastener. An estimated 80% increase in productively
- Developed a BOM sorting program that cut order prep and organization for a new machine from 2 days down to a few hours.
- Performed electrical wiring, soldering, and cable routing across multiple projects, contributing to automation improvements.

Projects

Inverse Kinematics–Controlled SCARA Plotter Robot (Dual Parallel Arm) — Fall 2025

- Designed and Built a belt-driven SCARA robot with dual parallel-link architecture
- Implemented closed-form inverse kinematics and real-time embedded motion control
- Validated accuracy by plotting precise geometric shapes with a Sharpie end-effector
- Designed a custom motor-control PCB and fully integrated hardware, electronics, and firmware

6x6x6 LED Cube — *Summer 2025*

- Designed and fabricated a custom PCB for a multiplexed LED cube display.
- Integrated soldering, wiring, circuit design, and programming to create a functional 3D light array.

Relevant Coursework

Motion Systems: Modeled and controlled complex dynamic systems through hands-on labs including Segway-style balancing robots and magnetic levitation using LabVIEW and Arduino.

Robotics: Explored mathematical modeling, kinematics, dynamics, and control of robotic manipulators and mobile robots, with practical labs in trajectory planning, sensor integration, and vision-based control.

Leadership and Activities

Invention Studio (Georgia Tech) | Atlanta, GA

Master of Electronics Tool Group , Director of Programming, Prototyping Instructor, Shift Manager — *August 2023 – Present*

- Promoted to Master of Electrical Tools, leading tool training, mentoring apprentices, and assisting students with advanced electrical projects.
- Serve as a leader of the makerspace, guiding technical decisions and contributing to operational improvements.
- Orchestrated and taught project-based workshops, emphasizing hands-on engineering education and tool safety.

Robojackets | Member — *August 2023 – May 2025*

Fabricated and machined components for a battlebots, including metalworking, electrical wiring, and mechanical assembly; iterated designs and competed in team competitions.